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EDWARDS EXT SERIES INSTRUCTION MANUAL



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PRX purge-restrictor
VRX vent-restrictor
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B753-03-880
Issue F Original

Instruction Manual

EXT Compound Molecular Pumps:
EXT255H, EXT255Hi and EXT255HVi



Description	Item Number
EXT255Hi/ISO100	B753-03-000
EXT255H/100CF	B753-02-000
EXT255H/ISO100	B753-01-000
EXT255HVi/ISO100	B753-07-000

EDWARDS
www.edwardsvacuum.com

Declaration of Conformity

We, Edwards,
Manor Royal,
Crawley,
West Sussex RH10 9LW, UK

declare under our sole responsibility that the product(s)

EXT Compound Molecular Pumps:

EXT255H/ISO100	B753-01-000
EXT255H/100CF	B753-02-000
EXT255Hi/ISO100	B753-03-000/B753-04-000
EXT255H/ISO100 Port variant	B753-06-000/B753-08-000
EXT255Hvi (Long)	B753-07-000

to which this declaration relates is in conformity with the following standard(s) or other normative document(s):

EN ISO12100-2: 2003	Safety of machinery. Basic concepts, general principals for design. Technical principals.
EN61010-1: 2001	safety requirements for electrical equipment for measurement, Control and laboratory use. General requirements.*
EN1012-2: 1997	Compressors and vacuum pumps. Safety requirements. Vacuum pumps.
EN61326: 1997† +A1: 1998 + A2: 2001 (Industrial Location, Class B Emissions)	Electrical equipment for measurement, control and laboratory use. EMC requirements.

* The pumps comply with EN61010-1 (2001) when installed in accordance with the instruction manual supplied with the pumps.

† EN61326 Immunity classification is dependent on the controller (see the controller manual for details).

following the provisions of:

2006/95/EC	Low Voltage Directive.
2004/108/EC	Electromagnetic Compatibility Directive.
98/37/EC	Machinery Safety Directive.

B. Brewster

Barrie Brewster, Technical Manager - Burgess Hill Products

24th October 2007
BURGESS HILL
Date and Place

This product has been manufactured under a quality system registered to ISO9001

P200-00-360 Issue C

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Associated publications

Publication title	Publication number
EXC Turbomolecular Pump Controllers: EXC100E and EXC100L	D396-20-880
E2M0.7 Rotary Pump	A371-31-880
RV Rotary Vane Pumps	A652-01-880

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1 Introduction

1.1 Scope and definitions

This manual provides installation, operation and maintenance instructions for the Edwards EXT255H, EXT255Hi and EXT255HVi Compound Molecular Pumps. You must use the Pumps as specified in this manual.

The EXT Compound Molecular Pumps are designed for use with an Edwards EXC Controller. Read this manual and the instruction manual supplied with your EXC Controller before you attempt to install or operate the equipment. The EXC Controller contains details of how to set up a pumping system and how to control accessories such as an air-cooler, vent-valve and bakeout band.

Important safety information in this manual is highlighted as WARNING and CAUTION instructions; you must obey these instructions. The use of WARNINGS and CAUTIONS is defined below.

WARNING



Warnings are given where failure to observe the instruction could result in injury or death to people.

CAUTION

Cautions are given where failure to observe the instruction could result in damage to the equipment, associated equipment and process

In accordance with the recommendations of EN61010, the following warning symbols may appear on the pump or its accessories:



Warning - refer to accompanying documentation.



Warning - risk of electric shock.



Warning - hot surfaces.



Protective conducter terminal.

The units used throughout this manual conform to the SI international system of units of measurement. Also throughout this manual, wherever flow rates are specified, the abbreviation 'sccm' is used to mean 'standard $cm^3\ min^{-1}$: this is a flow of 1 $cm^3\ min^{-1}$ at an ambient temperature of 0 °C and a pressure of 1013 mbar (1.013×10^5 Pa).

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1.2 General description

The EXT255H, EXT255Hi and EXT255HVi pumps are compound vacuum pumps that combine a turbomolecular pump and a drag pump on a single shaft. Refer to Figure 1 and 2, which show the internal structures of the EXT255H and EXT255HVi pumps.

The turbomolecular pump is made from a multi-stage axial-flow turbine which is optimised for operation in molecular flow conditions. The drag pump is made from pumping channels formed by helical grooves. The drag pump is designed for operation at higher pressures than conventional turbomolecular pumps.

The light alloy rotor (13) is machined from one piece to form the angled blades of the turbomolecular pump. The rotor blades rotate between angled stator blades (12) which are supported by spacer rings (11). The rotor and stator blades have an open structure at the pump inlet, and a more closed structure at the outlet. This configuration gives an optimum combination of pumping speed and compression when pumping gases of both high and low molecular weight. For the blades to be effective, their speed must be close to the thermal velocity of the gas. The rotor therefore rotates at up to 60000 r min^{-1}.

Gas from the inlet is compressed through each of the bladed stages of the turbomolecular pump and transferred to the drag pump. In the drag pump stages, the gas is further compressed as it is pumped around the channels formed by the helical grooves.

The rotating assembly is supported at the inlet end by a frictionless magnetic bearing (3) and by a precision ceramic ball bearing (8) at the outlet end. The ball bearing is lubricated from an oil reservoir and wick mechanism (9).

All of the pumps are supplied with an inlet-screen (2) fitted in the bore of the inlet-flange. The EXT255Hi pumps are also supplied with an inlet-strainer which fits in the interstage-port. The inlet-screen and inlet-strainer protect the pump against damage which would be caused by debris which enters the pump. The inlet-screen also protects you from the sharp blades in the pump.

All of the pumps have a vent-port you can use to vent the pump and your vacuum system to atmospheric pressure. The vent-port introduces vent gas part way up the pump rotor; this ensures maximum cleanliness. The pump is supplied with a manual vent-valve fitted to the vent-port. As described in Section 3.4, you can replace the manual vent-valve with a TAV5 or TAV6 solenoid-operated vent-valve (available as an accessory: see Section 7).

All of the pumps have a purge-port (Figure 4 or 5, item 3) in the motor and bearing housing chamber. In order to protect the bearing lubricant from the effects of high oxygen concentrations, you can introduce an inert purge gas through the purge-port. You can fit an optional vent-port adaptor and purge restrictor to the purge-port to control the flow rate of the purge gas and to filter the gas supply (see Section 7).

The EXT255Hi pump has an interstage-port, between the turbomolecular and drag pump. You can use the interstage-port to pump a separate chamber of your vacuum system or to back another turbomolecular pump. If the pump operates with an outlet pressure of less than 5 mbar (500 Pa) and an interstage inlet flow of less than 70 sccm (1.17 mbar l s^{-1}, 117 Pa l s^{-1}), the vacuum performance at the turbomolecular pump-inlet is independent of the interstage operating conditions.

The EXT255HVi pump also has an interstage-port (14). The interstage-port is located in the turbomolecular pumping region of the pump and, together with the higher conductance provided by the larger size interstage-port, enables higher interstage pumping speeds than the EXT255Hi.

Electrical connection between the EXT and the EXC Controller is by a 19-way connector and a pump-to-controller cable. The cable is a separate item and is available in a choice of lengths (see Section 7 for details). If you use an EXC100L controller it is not necessary to buy a separate cable.

The pump may be cooled using air-cooled or water-cooled optional accessories. Refer to Section 3 for guidance on applications and cooling requirements. Pumps with a Conflat flange are supplied with a water-cooler.

All EXT pumps have thermal sensors to monitor the motor and pump-body temperature.

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1.3 Vent options and vent control

To maintain the cleanliness of your vacuum system, we recommend that, whenever you switch the pump off, you vent the pump (or vacuum system) when the speed of the EXT pump is between full rotational speed and 50% of full rotational speed. At and above 50% of full rotational speed, the rotor spins fast enough to suppress any backstreaming of hydrocarbon oil from your backing pump.

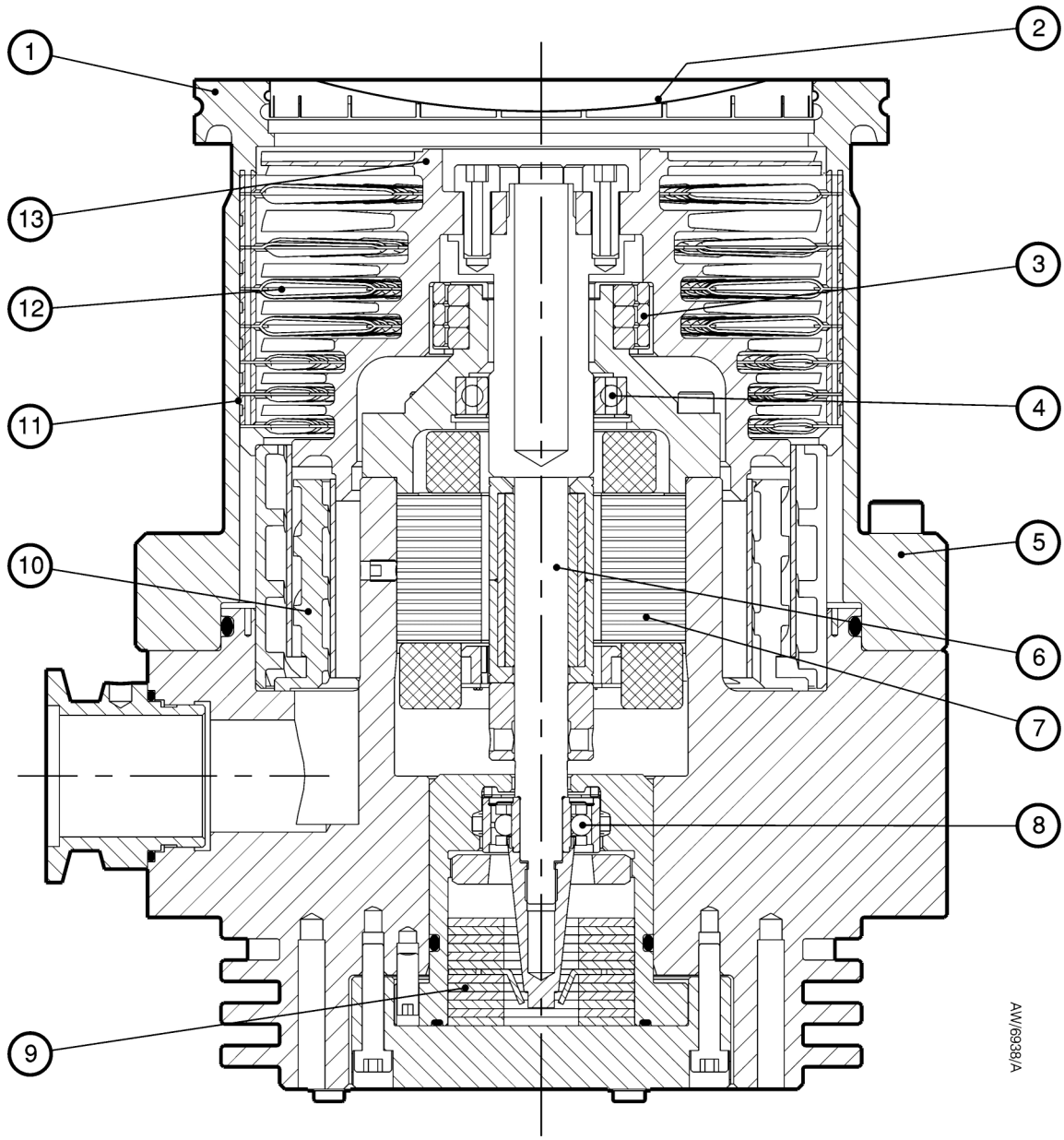
However, if you vent the pump when it is at full rotational speed and the rate of pressure rise is too high, the pump life may be reduced. We therefore recommend that you either limit the vent rate (see Section 2.3), or only open the vent-valve after the EXT pump speed has fallen to 50% of full rotational speed.

The rate of pressure rise cannot be controlled by the manual vent-valve, so if you use the manual vent-valve, you must only open the vent-valve after the EXT pump speed has fallen to 50% of full rotational speed.

If you use a TAV5 vent-valve, but you cannot limit the rate of pressure rise, you must only open the vent-valve after the speed of the EXT pump has fallen to 50% of full rotational speed. If you use the EXC Controller to control your TAV5 vent-valve, configure the EXC Controller to select this option: refer to Section 3.4 for more information. The EXC Controller is factory set to vent when the EXT pump is at 50% of full rotational speed after you select Stop.

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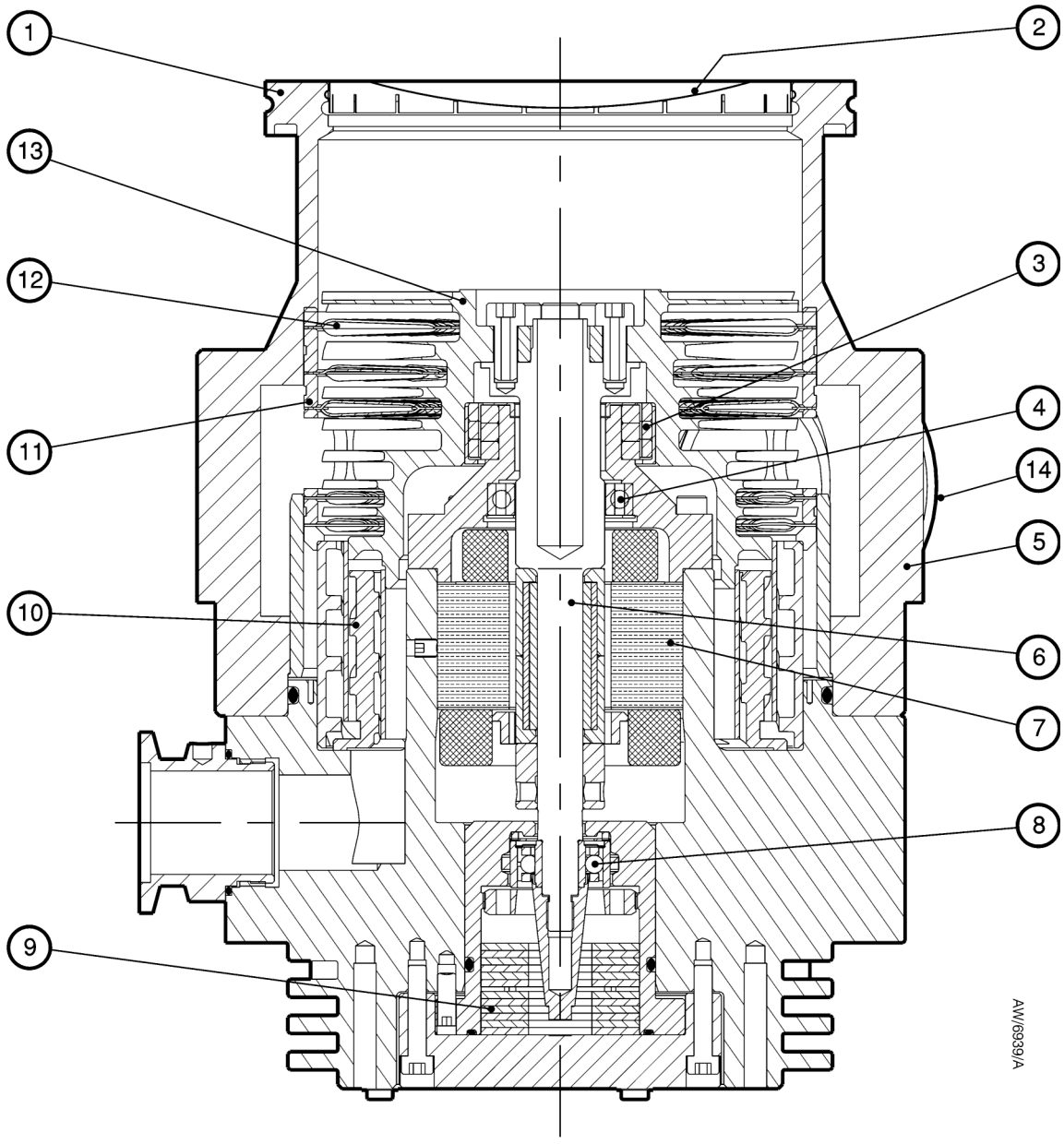
Figure 1 - Cross-section view of EXT255H pump



1. Inlet-flange
2. Inlet-screen
3. Magnetic bearing
4. Safety bearing
5. Envelope
6. Shaft
7. DC motor
8. Ball bearing
9. Oil reservoir
10. Drag stage
11. Spacer ring
12. Stator
13. Rotor

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Figure 2 - Cross-section view of EXT255HVi pump



1. Inlet-flange
2. Inlet-screen
3. Magnetic bearing
4. Safety bearing
5. Envelope
6. Shaft
7. DC motor
8. Ball bearing
9. Oil reservoir
10. Drag stage
11. Spacer ring
12. Stator
13. Rotor
14. Interstage-port

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2 Technical data

2.1 General

Performance	See Table 1
Dimensions	See Figure 4 and 5
Maximum inlet-flange temperature	100 °C
Maximum magnetic field	3.5 mT Horizontal field >7mT Vertical field
Installation category	EN61010 part 1, Category 1
Pollution degree	CAN/CSA, UL, EN61010 part 1
Altitude	2000 m
Humidity	10 to 90% non condensing
Electrical supply	80 V d.c.
Equipment type	Fixed equipment, for indoor use only

2.2 Pumping media

WARNING

Vent dangerous gases and gas mixtures safely. Do not expose people to these gases.

WARNING

Do not use EXT pumps to pump explosive gas mixtures as the pumps are not suitable for this purpose.

WARNING

On EXT255Hi and EXT255HVi pumps, gas pumped through the interstage-port will mix with gas pumped through the pump-inlet. Ensure that the gases will not react or combine to form dangerous gases and substances.

CAUTION

Do not use an EXT to pump gases containing more than 20% oxygen unless the pump is gas purged. If you do, the lubricant will polymerise and the pump will fail prematurely.

Note: Note that concentrations of gases may be modified by the compression of the pump.

2.2.1 Pumps without gas purge

The pumps are designed to pump the following residual gases normally used in high-vacuum systems:

- Air
- Nitrogen
- Carbon dioxide
- Carbon monoxide
- Krypton
- Helium
- Neon
- Argon
- Hydrogen
- Ethane
- Propane
- Butane
- Methane

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You can use the pumps to pump oxygen and water vapour, subject to the following conditions:

- Oxygen — The oxygen concentration must be less than 20% by volume.
- Water vapour — You must ensure that vapour does not condense inside the pump; refer to Section 3.7.3.

If you wish to pump a gas not in the list above, contact your supplier for advice. If you do not contact your supplier, you may invalidate the warranty on the pump. The pumps are not suitable for pumping aggressive or corrosive gases.

2.2.2 Pumps with gas purge

When purged with an inert gas, the pumps can be used to pump oxygen in concentrations above 20% by volume.

2.3 Vent gas specification and vent control data

Although the pump may be vented to atmospheric air, high relative humidity of the air may greatly increase the subsequent pump-down time. To reduce pump-down times you should vent the pump with dry, clean gases.

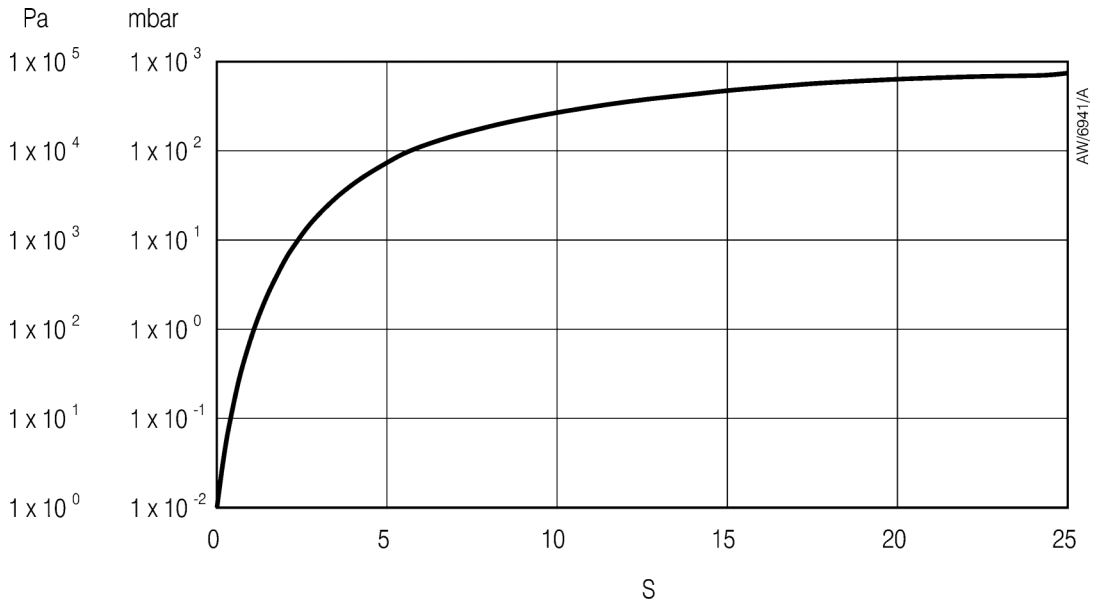
Vent gas	Dry air, nitrogen, argon or other inert gases
Maximum dew point at atmospheric pressure	- 22 °C
Maximum size of particulates	1 µm
Maximum concentration of oil	0.1 parts per million
Time for rotational speed to reach 50%	>15 sec
Rate of pressure rise	see Figure 3

2.4 Purge gas specification

Purge gas	Dry nitrogen, argon or other inert gases
Maximum dew point at atmospheric pressure	-22 °C
Maximum size of particulates	1 µm
Maximum concentration of oil	0.1 parts per million
Allowable purge gas flow (when required)	20 to 100 sccm (0.33 to 1.67 mbar l s^{-1}, 33 to 167 Pa l s^{-1})
Recommended purge gas flow	25 sccm (0.42 mbar l s^{-1}, 42 Pa l s^{-1})
Maximum allowable purge gas supply pressure	2 bar (gauge) ; 29 psig, 3 x 10^5 Pa

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Figure 3 - Maximum allowed rise in backing pressure against time for controlled venting: system pressure (mbar/Pa) against time (s)



2.5 Cooling-water

The following cooling water specification corresponds to a typical high-quality drinking water specification. Check with your water supply authority if you are in doubt about the quality of your supply.

Quality	Mechanically clean and optically clear with no deposits or turbidity.
pH value	6.0 to 8.0
Maximum calcium carbonate concentration	75 parts per million
Maximum chloride concentration	100 parts per million
Minimum oxygen concentration	4 parts per million
Minimum water-cooling flow rate (at 15 °C)	15 l h^{-1}
Water temperature	See Table 1
Maximum water pressure	5 bar (gauge), 73.5 psig, 6 x 10^5 Pa
Materials exposed to cooling water	Nickel plated brass

Table 1 - Technical data

Parameter	EXT255H	EXT255Hi	EXT255HVi	Notes
Mass				
DN100ISO-K inlet flange	5.6 kg	5.6 kg	6.5 kg	
DN100CF inlet flange	8.2 kg	N/A	N/A	
Inlet-flange	DN100ISO-K/ DN100CF	DN100ISO-K	DN100ISO-K	
Outlet-flange	DN25NW	DN25NW	DN25NW	
Vent-port	1/8 inch BSP	1/8 inch BSP	1/8 inch BSP	
Interstage-port	-	DN25NW	DN40NW	
Purge-flange	DN10NW*	DN10NW*	DN10NW*	

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Table 1 - Technical data (continued)

Parameter	EXT255H	EXT255Hi	EXT255HVi	Notes
Inlet pumping speed				
N_2 † ‡	220 l s^{-1}	220 l s^{-1}	195 l s^{-1}	Pb < 5 mbar (500 Pa) Qi < 70 sccm (1.17 mbar l s^{-1}) (117 Pa l s^{-1})
He † ‡	230 l s^{-1}	230 l s^{-1}	155 l s^{-1}	Pb < 1 mbar (100 Pa) Qi = 0 sccm
H_2 † ‡	180 l s^{-1}	180 l s^{-1}	100 l s^{-1}	Pb < 0.5 mbar (50 Pa) Qi = 0 sccm
Inlet compression ratio				
N_2 † ‡	> 1 x 10^8	>1 x 10^8	3 x 10^8	Pb < 5 mbar (500 Pa)
He † ‡	4 x 10^5	2.5 x 10^5	2 x 10^5	Pb < 2 mbar (200 Pa)
H_2 † ‡	1.6 x 10^4	1.1 x 10^4	5 x 10^3	Pb < 1 x 10^{-2} mbar (1 Pa)
Interstage pumping speed ‡				
N_2 †	-	10 l s^{-1}	21 l s^{-1}	Pb = 5 mbar (500 Pa) Qi = 240 sccm (4 mbar l s^{-1}) (400 Pa l s^{-1}) Qp = 0 sccm
He†	-	10 l s^{-1}	23 l s^{-1}	Pb = 5 mbar (500 Pa) Qi = 100 sccm (1.67 mbar l s^{-1})(167 Pa l s^{-1}) Qp=<20 sccm (0.33 mbar l s^{-1})(33 Pa l s^{-1})
Ultimate pressure with rotary vane backing pump **				
DN100ISOK inlet flange	< 5 x 10^{-9}	< 5 x 10^{-9}	< 5 x 10^{-9}	
DN100CF inlet flange	< 5 x 10^{-10}	-	-	
Ultimate pressure with diaphragm backing pump ††				
DN100ISOK inlet flange	< 5 x 10^{-8}	< 5 x 10^{-8}	< 5 x 10^{-8}	
DN100CF inlet flange	< 5 x 10^{-8}	-	-	
Maximum backing pressure ‡‡				
N_2	12 mbar (1200 Pa)	12 mbar (1200 Pa)	12 mbar (1200 Pa)	
He	10 mbar (1000 Pa)	10 mbar (1000 Pa)	10 mbar (1000 Pa)	
H_2	3 mbar (300 Pa)	3 mbar (300 Pa)	3 mbar (300 Pa)	
Minimum backing pump displacement	0.6 m^3 h^{-1}	0.6 m^3 h^{-1}	0.6 m^3 h^{-1}	
Maximum continuous inlet pressure ***				
water-cooling at 15 °C ***	2 x 10^{-1} mbar, 20 Pa	2 x 10^{-1} mbar, 20 Pa	2 x 10^{-1} mbar, 20 Pa	
air-cooling at 35 °C***	2 x 10^{-2} mbar, 2 Pa	2 x 10^{-2} mbar, 2 Pa	2 x 10^{-2} mbar, 2 Pa	

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Table 1 - Technical data (continued)

Parameter	EXT255H	EXT255Hi	EXT255HVi	Notes
Operating attitude	Vertical and upright through to horizontal ±2°			
Nominal rotational speed	60000 r min^{-1}	60000 r min^{-1}	60000 r min^{-1}	
Standby rotational speed	42000 r min^{-1}	42000 r min^{-1}	42000 r min^{-1}	
Starting time to 90% speed				
EXC100E	190 sec	190 sec	190 sec	
EXC120	130 sec	130 sec	130 sec	
EXC300	100 sec	100 sec	100 sec	
Cooling method	Forced-air/ water	Forced-air/ water	Forced-air/ water	
Ambient air temperature (forced-air cooling)	0 - 35 °C	0 - 35 °C	0 - 35 °C	
Water temperature (water-cooling)	10 - 20 °C	10 - 20 °C	10 - 20 °C	
Noise level (at 1 metre)	< 50 dB(A)	< 50 dB(A)	< 50 dB(A)	
Recommended controller	EXC120	EXC120	EXC120	
EXC120 maximum VA input	250 VA	250 VA	250 VA	with bakeout band
Quiescent power	25W	25W	25W	
Recommended backing pump †††	RV3	RV3	RV3	

* *Not supplied - pump supplied with port blanked.*

† *Pb = backing pressure, Qi = flow through the interstage-port (EXT255Hi/EXT255HVi only), Qp = flow through pump-inlet.*

‡ *Pumping speeds are without inlet-screen or inlet-strainer (EXT255Hi/EXT255HVi only). Inlet-screens and inlet-strainers reduce speed by approximately 10%.*

** *Ultimate pressure 48 hours after bakeout with Pb < 1 x 10^{-2} mbar (1 Pa).*

†† *Ultimate pressure 48 hours after bakeout with Pb < 5 mbar (500 Pa).*

‡‡ *Inlet pressure has risen to 1 x 10^{-3} mbar (1 x 10^{-1} Pa).*

*** *Above this pressure, rotational speed drops below nominal.*

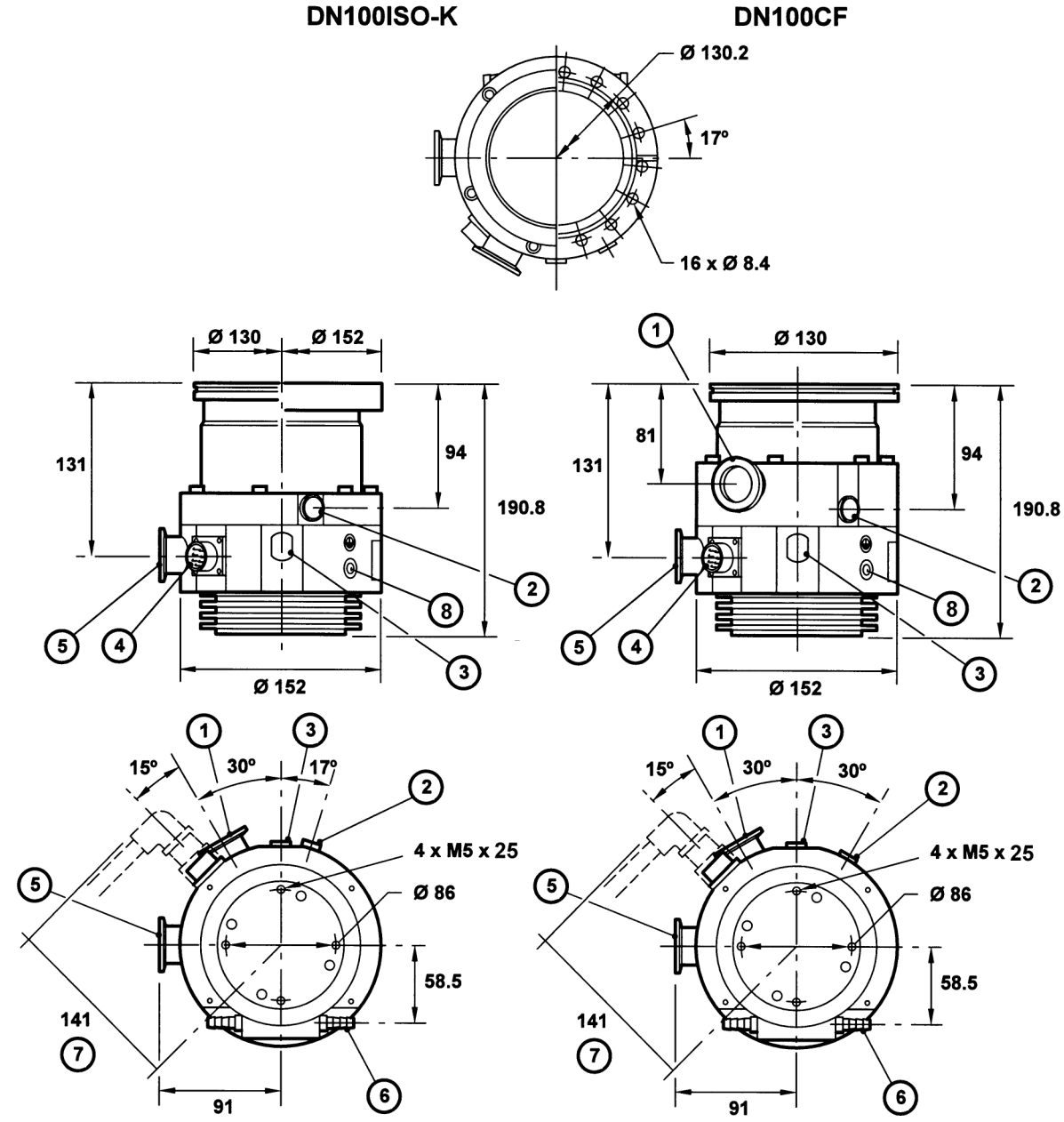
††† *A larger backing pump may be required for maximum throughput.*

2.6 Materials exposed to gases pumped

The following materials and component types are exposed to the gases pumped: aluminium alloys, stainless steels, fluoroelastomer and nitrile 'O' rings, hydrocarbon lubricant, felt, rare earth magnets, silicon nitride, phenolic resin and carbon-fibre reinforced epoxy resin.

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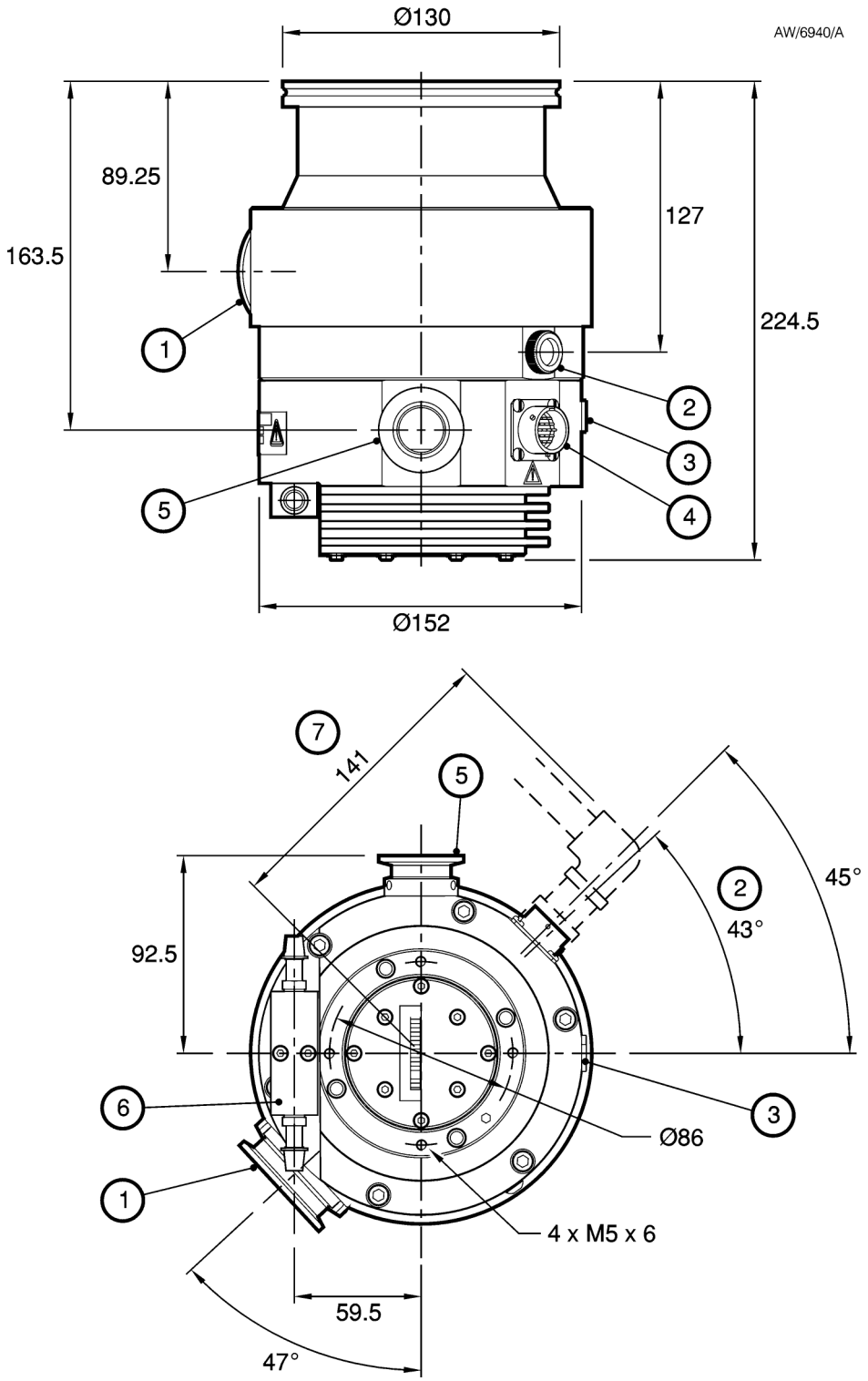
Figure 4 - EXT255H/EXT255Hi dimensions (mm)



1. Interstage-port (EXT255Hi only)
2. Vent-valve
3. Purge-port
4. Electrical supply connector
5. Backing-port
6. Cooling-water connectors
7. Allowance for right-angle cable connector
8. Earth (ground) bonding point

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Figure 5 - EXT255HVi dimensions (mm)



1. Interstage-port
2. Vent-valve
3. Purge-port
4. Electrical supply connector
5. Backing-port
6. Cooling-water connectors
7. Allowance for right-angle cable connector

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3 Installation

WARNING



Safely route all vacuum, vent/purge gas and cooling-water pipelines, and all electrical cables and wires, so that people cannot trip over them.

3.1 Unpack and inspect

The pump is packed to prevent damage in transit. Take care when you unpack the pump to avoid excessive shocks which could damage the bearings and reduce the life of the pump. The pump is supplied with the inlet and outlet sealed to prevent entry of dust and vapour. Do not remove these seals until you are ready to install the pump on your vacuum system.

Remove all packing materials and check the pump. If the pump is damaged, notify your supplier and the carrier in writing within three days; state the Item Number of the pump together with your order number and your supplier's invoice number. Retain all packing materials for inspection. Do not use the pump if it is damaged.

Check that your package contains the items listed in Table 2. If any of these items is missing, notify your supplier in writing within three days.

Table 2 - Checklist of items

Quantity	Description	Check (✓)
1	EXT turbomolecular pump	❒
1	Inlet seal (trapped 'O' ring, Co-Seal or copper compression gasket)	❒
1	Inlet-strainer (EXT255Hi only, fitted in the interstage-port)	❒
1	Interstage seal (EXH255HVi only, fitted in the interstage-port)	❒

If the pump is not to be used immediately, store the pump in suitable conditions as described in Section 6.1.

Do not discard the packing materials; retain them to repack the pump when you return it for service.

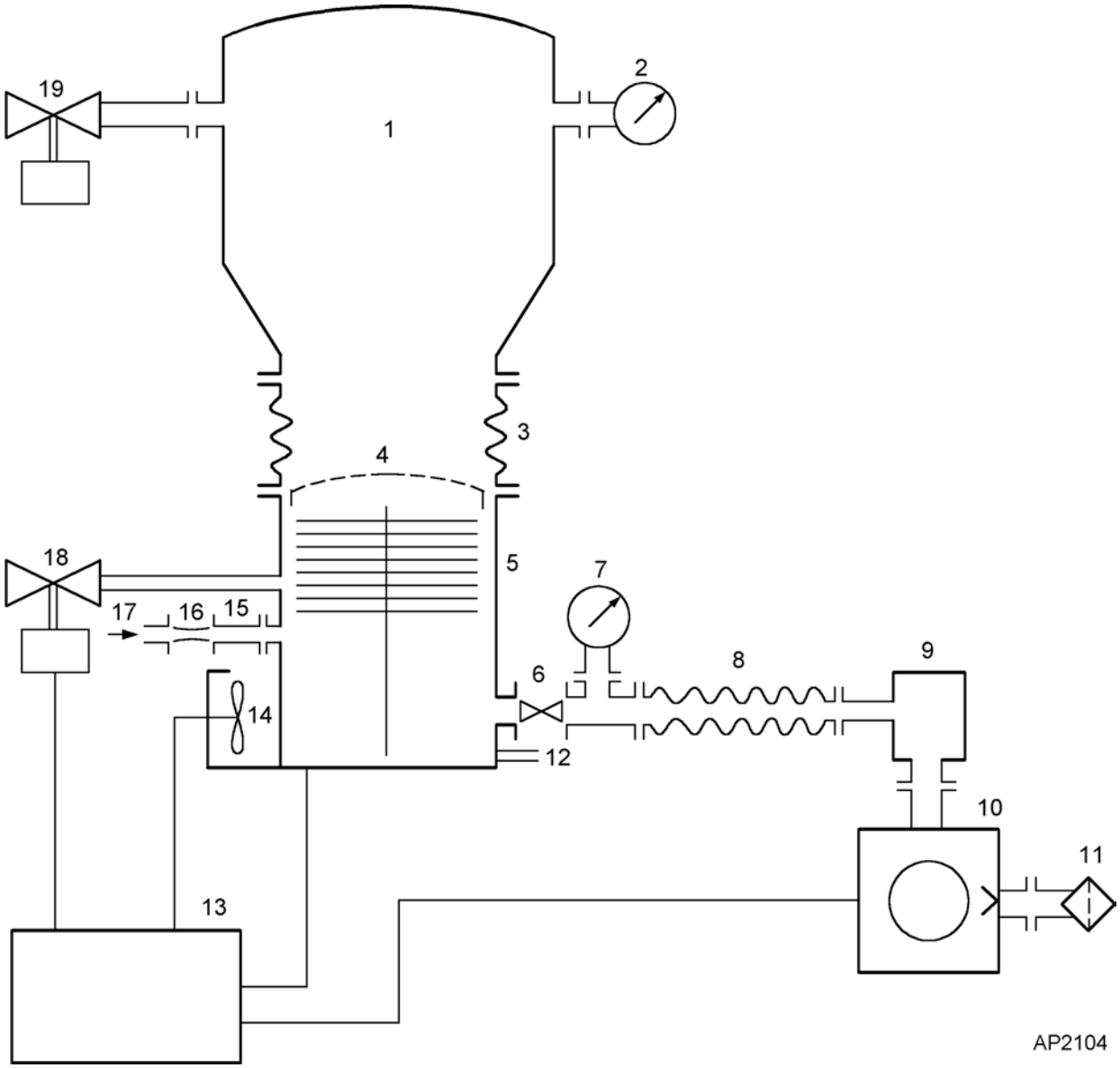
3.2 Typical installation

A typical pumping system with an EXT255H pump is shown in Figure 6. A typical pumping system with an EXT255Hi or EXT255HVi pump is shown in Figure 7. When necessary, purge the EXT pump with inert gas as described in Section 3.5.

The accessories available for these EXT pumps are detailed in Section 7; the accessories are shown in Figure 9.

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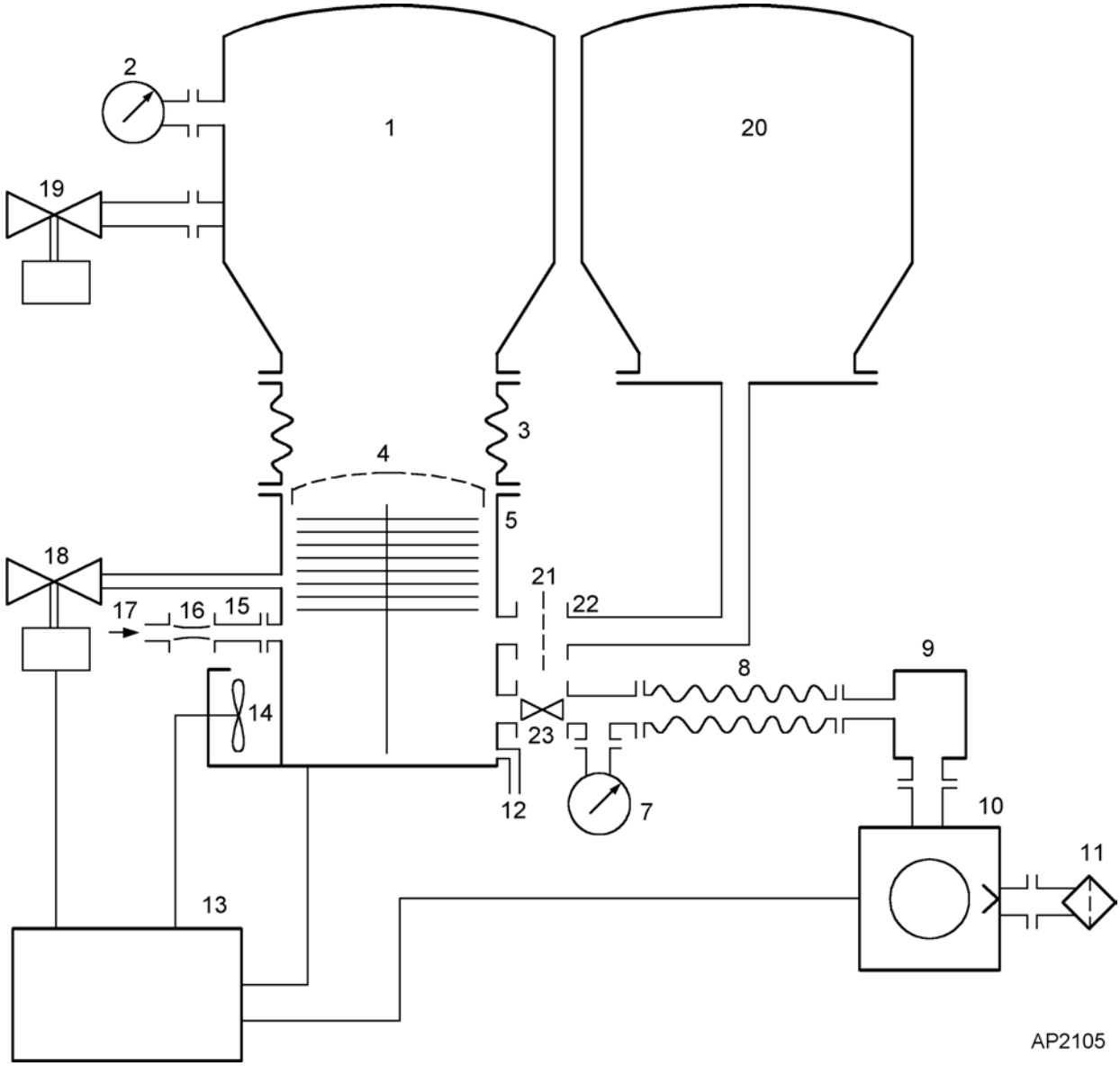
Figure 6 - Typical pumping system with an EXT255H pump



1. Vacuum system
2. High-vacuum gauge
3. Vibration isolator
4. Inlet-screen
5. EXT pump
6. Backing valve
7. Vacuum
8. Flexible bellows
9. Foreline trap
10. Rotary backing-pump
11. Mist filter
12. Cooling-water connectors
13. EXC controller
14. Air-cooler
15. Vent port adaptor
16. PRX10 purge restrictor
17. Regulated purge gas supply
18. Vent-valve
19. Alternative position for vent-valve

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Figure 7 - Typical pumping system with an EXT255Hi/EXT255HVi pump



1. Vacuum chamber 1
2. High-vacuum gauge
3. Vibration isolator
4. Inlet-screen
5. EXT pump
6. Not used
7. Vacuum gauge
8. Flexible bellows
9. Foreline trap
10. Rotary backing-pump
11. Mist filter
12. Cooling-water connectors
13. EXC controller
14. Air-cooler
15. Vent port adaptor
16. PRX10 purge restrictor
17. Regulated purge gas supply
18. Vent-valve
19. Alternative position for vent-valve
20. Vacuum chamber 2
21. Inlet-strainer
22. Interstage-port
23. Backing valve

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3.3 Connect to the vacuum system

WARNING

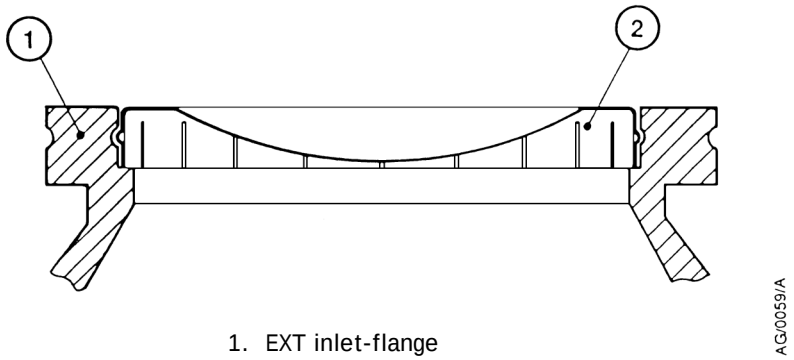
Install the pump in the vacuum system before you connect the EXC Controller. This will ensure that the pump cannot operate and injure people during installation.

3.3.1 Inlet screen

Do not remove the inlet-screen unless you can be sure that there is no danger that debris can fall into the pump. In order to avoid the danger of injury from the rotor blades, do not remove the inlet-screen until you are ready to mount the pump onto your system. If the screen is removed, the pumping speed will increase by approximately 10%.

To remove the inlet-screen carefully extract it from the inlet-flange using a bent wire hook. To replace a screen which has been removed, install it as shown in Figure 8, with the Edwards logo uppermost. Ensure that the dimples on the rim of the screen engage in the groove in the pump flange. If necessary, gently bend the tabs of the screen outwards to ensure a tight fit.

Figure 8 - Correct installation of the inlet-screen



1. EXT inlet-flange
2. Inlet-screen

3.3.2 Mechanical fixing

WARNING

Do not operate the EXT pump until it is securely fixed. If the pump seizes, the stored energy of the rotor can cause rapid movement of the pump, which may cause damage and injury to people.

There are two ways in which the EXT pump can be securely fixed. Ideally, the EXT pump should be securely fixed by its inlet-flange to a rigid, firmly fixed vacuum system: see Section 3.3.4. If this is not possible because of the nature of the vacuum system, the base of the EXT pump must be securely fixed to a firm support: see Section 3.3.3.

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3.3.3 Base mounting

Ensure that the base of the pump is securely fixed to a firm support (refer to Figure 4 and 5 for the fixing hole details). If the pump supports the weight of the vacuum system, the mass of the vacuum system must be no more than 20 kg.

You must also ensure that your mounting method meets the following requirements, so that the EXT pump will remain secure in the event of a pump seizure:

- The support mounting must be able to withstand a destructive torque of 620 Nm.
- Fit cap-head fixing screws through the tapped fixing-holes in the base of the pump (see Figure 4 and 5): use M5 screws.
- The fixing screws must comply with ISO 898-1, with a strength class of 12.9 (nominal tensile strength 1200 MPa).
- The fixing screw engagement length must be 6 mm or more.
- Tighten the fixing screws to a torque of 12 Nm (1.22 kgf m).

3.3.4 Inlet connection and orientation

The EXT pump can be fixed to the vacuum system by the inlet-flange. The pump can be mounted in any attitude from vertical and upright through to horizontal (±2°). If the pump is mounted horizontally and you use a rotary vane pump to back the EXT pump, the backing port must point vertically downwards (±20°) to reduce the risk of contamination from the backing pump oil.

Make sure that the pump-inlet and all components fitted to the pump-inlet are clean and dust-free. If the pump-inlet is not kept clean, the pump-down time may be increased.

The inlet-connection of the EXT pump is a CF flange or an ISO flange:

- If the pump has a CF flange, use the copper compression gasket supplied with the pump and use a full complement of bolts to connect the inlet-flange of the pump to the vacuum system.
- If the pump has an ISO flange, use the Edwards trapped 'O' ring supplied with the pump and use a minimum of four claw clamps to connect the inlet-flange of the pump to the vacuum system. Ensure that each claw clamp is tightened to a torque of 10 Nm or more.

  Alternatively, use a rotatable collar and the trapped 'O' ring supplied with the pump to connect the inlet-flange of the pump to the vacuum system; use a full complement of bolts with the rotatable collar.

Ensure that no torque or other forces are transmitted to the pump from the vacuum system or the associated pipelines.

If necessary, fit an inlet vibration isolator between the pump-inlet and the vacuum system: refer to Section 7.4.9 for the Item Numbers, and refer to the instruction manual supplied with the vibration isolator for installation details. If you fit a vibration isolator, you must securely fix the base of the EXT pump as described in Section 3.3.3.

Note: The first time you pump down the system to vacuum, you must re-tighten the bolts which secure the inlet-flange.

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3.3.5 Backing connection

Use suitable vacuum tubing and connectors to connect the NW flange of the backing-port to your backing-pump. If necessary, use flexible pipe or bellows to reduce the transmission of vibration from the backing-pump to the EXT pump.

We recommend that you use an Edwards RV backing-pump. The backing-pump can also be controlled by the EXC Controller (EXC120 EXC300). The minimum size of the backing-pump required is given in Table 1. You may have to use a larger backing-pump if you run the pump at a high inlet pressure or high throughput, or if you purge the pump with more than 25 sccm (0.42 mbar l s^{-1}, 42 Pa l s^{-1}) of purge gas. The EXT255H is also suitable for use with diaphragm backing-pumps although the effect of high backing pressure on the pump's performance and cooling requirements should be noted (see Table 1 and Section 3.7).

Do not use the EXT pump with a backing pressure below 5 x 10^{-4} mbar (5 x 10^{-2} Pa). Lower backing pressures will increase the evaporation rate of the lubricating oil and so will reduce the life of the bearings.

3.3.6 Interstage connection (EXT255Hi/EXT255HVi only)

The EXT255HVi pump is supplied with an interstage-port seal, and the EXT255Hi pump is supplied with a combined inlet-strainer/interstage-port seal; leave the inlet-strainer in the interstage-port, unless you are sure that debris cannot be drawn into the interstage-port.

Use the seal supplied and suitable vacuum tube and connectors to connect the interstage-port to your vacuum system or to the outlet flange of another turbomolecular pump (if you use the EXT255Hi or EXT255HVi pump to back another turbomolecular pump).

3.4 Vent-valve connection and control

When you design your system and when you install a vent-valve, take note of the information in Section 1.3 and 2.3. You can vent the EXT pump and your vacuum system by any of the following methods:

- Use the manual vent-valve supplied. Take care not to open the manual vent-valve too quickly.
- Use a TAV5 or TAV6 solenoid vent-valve accessory (see Section 7) in place of the manual vent-valve.
- Use a TAV5 or TAV6 solenoid vent-valve connected to a convenient flange on your vacuum system.
- Use an alternative valve connected to your vacuum system.

If you use the manual vent-valve, you must open the vent-valve only after the EXT pump speed has fallen to 50% of full rotational speed.

Do not vent from the backing line. If you vent into your vacuum system, select a point upstream of the pump, to prevent oil backstreaming from the backing line.

If you use the TAV5 or TAV6 vent-valve we recommend that you control it from the EXC controller.

Table 3 gives an indication of the appropriate orifice size to be fitted to the vent valve for given vacuum system volumes in order that the vent rate is kept within the limits given in Section 2.3.

Table 3 - Vent-restrictor orifice diameter (with atmospheric pressure at the inlet of the vent-valve)

Vacuum system volume (l)	Orifice diameter (mm)
< 20	≤ 1.0
< 10	≤ 0.7
< 5	≤ 0.5
< 2	≤ 0.35

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3.5 Purge gas connection

3.5.1 Connect the purge gas

If you want to supply a purge gas to the pump, fit a vent port adaptor (see Section 7.4.7) in place of the blank plug (Figure 4 or 5, item 3). Connect your gas supply to the purge port.

You must limit the flow rate of the purge gas to the allowed range, also specified in Section 2.4. To limit the flow rate, use a flow controller or a pressure regulator and calibrated flow restrictor. The PRX10 purge restrictor accessory (see Section 7.3) is suitable for this purpose. Adjust the PRX10 as described in the instruction manual supplied with the accessory.

3.5.2 Recommended purge gas flow

The recommended purge gas flow for typical applications is 25 sccm (0.42 mbar l s^{-1}, 42 Pa l s^{-1}). This flow will protect the pump when you pump oxygen in concentrations above 20% by volume.

3.6 Electrical installation

Always make the electrical connections to the EXT pump after the pump has been installed on your vacuum system.

The EXT pump should be electrically bonded to earth (ground) using the connection provided (Figure 4 or 5, item 8).

The EXC Controller provides the electrical supply to the EXT pump through the multiway pump-to-controller cable. Connect and lock the bayonet-connectors at the ends of the cable to the mating connectors on the pump and the EXC Controller.

The EXC Controller is designed to allow a pumping system to be configured in a variety of ways, from a basic manually-operated system to a fully automatic system with remote control. Refer to the instruction manual supplied with the EXC Controller to complete the electrical installation.

3.7 Cooling

3.7.1 Introduction

CAUTION

You must cool the pump by forced-air or water-cooling to prevent damage to the bearing lubricant.

We recommend that wherever possible, you cool the pump by forced-air or water-cooling.

You must use water-cooling in any of the following operating conditions:

- Backing pressure >10 mbar (1000 Pa).
- Backing pressure > 8 mbar (800 Pa) and interstage flow > 30 sccm (0.5 mbar l s^{-1}, 50 Pa l s^{-1}).
- Backing pressure > 5 mbar (500 Pa) and interstage flow > 80 sccm (1.3 mbar l s^{-1}, 130 Pa l s^{-1}).
- Ambient temperature > 35 °C.
- When you use a bakeout band.

In all other operating conditions, you can use forced-air cooling. If you use forced-air to cool the pump, you must ensure that there is an adequate supply of cooling-air to the pump.

During operation, if the temperature of any surface of the pump is higher than 50 °C, the pump is too hot and you must increase the cooling.

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3.7.2 Forced-air cooling

An air-cooler accessory is available for the EXT pumps (refer to Section 7). Fit the air-cooler as described in the instruction manual supplied with it. If you wish to use an alternative fan for air-cooling, ensure that the flow rate is above 70 $m^3 h^{-1}$ (40 cfm)

3.7.3 Water-cooling

A water-cooler accessory is available for the EXT pumps (refer to Section 7.3.4). Fit the water-cooler as described in the instruction manual supplied with it.

The cooling-water supply must comply with the specification given in Section 2.5. Pipes in the water-cooling circuit may become blocked if the cooling-water contains too much calcium carbonate or if it contains particulates which are too large. Corrosion of the water-cooling circuit may occur if there is too little calcium carbonate and oxygen in the water. Good quality drinking water is usually suitable for water-cooling. If in doubt, you must check the quality of your cooling-water supply and, if necessary, provide treatment and filtration.

Connect the cooling-water supply to the water-cooler on the pump as described below. Either of the two riffled connectors on the water-cooler can be used for the water supply or return connections.

1. Push reinforced hose (approximately 6 mm internal diameter) over the ends of the riffled hose connectors on the water-cooler on the pump.

2. Attach the hose with strong hose clips and make sure that they are tightened securely.

Alternatively, unscrew the riffled hose connectors and remove them from the water-cooler and make direct connections to the 1/8 BSP female threaded fittings on the water-cooler.

You must turn off the cooling-water supply when you switch off the pump to prevent condensation of vapours inside the pump. The EXC Controller (EXC120 and EXC300) can operate a solenoid-valve for this purpose.

If you want to remove the pump for maintenance, and you do not want to break the cooling-water circuit, unscrew the two M4 cap-head fixing-screws and remove the water-cooler from the pump. Make sure that there is a layer of thermal contact grease on the water-cooler before you refit it to the pump.

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4 Operation

WARNING



Do not disconnect the pump-to-controller cable when the EXT pump is operating. If you do, there may be a risk of injury or death by electric shock.

WARNING



Do not operate the pump unless it is connected to your vacuum system. If you do, the pump rotor can cause injury. The pump rotor rotates at very high speeds and you may not be able to see that the pump is rotating.

WARNING



Do not expose any part of your body to vacuum. If you do, you may be injured.

4.1 Start-up

Use the procedure below to start up a basic, manually-controlled pumping system with a manual vent-valve and an end user version EXC Controller (for example, EXC120 or EXC300). Refer to the EXC instruction manual where the backing pump and accessories are automatically controlled by the EXC Controller.

1. Turn the manual vent-valve clockwise to close it.
2. Turn on the cooling-water supply (if water-cooling is used).
3. Start the backing-pump.
4. When the vacuum system pressure is approximately 10 mbar (1×10^3 Pa) or less, press the Start/Stop button on the EXC Controller to start the EXT pump.
5. The pump will then accelerate to full operating speed. The pump rotational speed is shown by the LED speed indicator on the front panel of the EXC Controller.

Note: The first time you pump down the system to vacuum, you must re-tighten the bolts which secure the inlet-flange: refer to Section 3.3.4.

4.2 Stand-by

You can press the Standby button on the EXC Controller to operate the EXT pump at reduced rotational speed. Select Standby before or after Start-up, for any of the following reasons:

- To extend pump-bearing life and still maintain adequate vacuum pumping performance (for example, when you leave a system under vacuum over holiday periods).
- To increase system pressure or to extend the maximum inlet pressure range of the pump where this suits a particular process.

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4.3 Shut-down

Note: In an emergency only, open the vent-valve quickly to decelerate the pump rotor in the shortest possible time.

Use the procedure below to shut down a basic, manually-controlled pumping system with a manual vent-valve and an end user version EXC Controller (for example, EXC120 or EXC300). Refer to the EXC Controller instruction manual where the backing-pump and accessories are automatically controlled by the EXC Controller.

1. Close the valve in the backing-line connecting the EXT pump to the backing-pump.
2. Switch off the backing-pump.
3. Press the Start/Stop button on the EXC Controller to switch off the EXT pump.
4. When the EXT pump rotational speed has fallen to below 50% of full rotational speed, turn the manual vent-valve anticlockwise to open it. Ensure that the rate of pressure rise does not exceed the allowed rate of pressure rise, otherwise you can damage the pump: refer to Section 1.3 and 2.3.
5. If water-cooling is in use, turn off the cooling-water supply.

4.4 Safety interlocks and control system

The pump protection and safety interlock features are listed below. Refer to the instruction manual supplied with the EXC Controller for a full description of these features (note that references to LEDs and buttons do not apply to OEM versions of the EXC controller):

- The EXC Controller monitors the temperature of the EXT pump and the electrical power consumption of the pump. If the EXC Controller detects excessive power consumption or temperature, the rotational speed of the pump motor is reduced until the power and temperature return to normal
- If the rotational speed is reduced to 50% of nominal speed, then the pump is stopped immediately (or after a user defined time delay) and the Fail LED on the EXC Controller lights
- If pump rotational overspeed is detected by the EXC Controller, the pump is stopped immediately and the FAIL LED on the EXC Controller lights.

If the Fail LED lights, switch off the backing-pump immediately and vent the EXT pump. Once the EXT pump has stopped, rectify the cause of the failure (refer to Section 5.5), press the EXC Controller Start/Stop button to reset the Fail condition, and restart the EXT pump. If the pump is hot, allow sufficient time for it to cool before you restart it.

4.5 Bakeout

CAUTION

When you bake the EXT pump to above 70 °C at the inlet-flange, you must cool the pump by water-cooling, to prevent damage to the bearing lubricant.

If you heat your EXT pump (and your vacuum system), you will speed up the degassing process so that the pump will reach ultimate vacuum in the shortest possible time. If you heat the pump, this will also prevent condensation of vapours inside the pump.

You can use the Edwards BX bakeout band to heat the pump (refer to Section 7). Fit the band around the pump, just below the inlet-flange. When you bake the pump or the system, make sure that the temperature of the inlet-flange does not exceed 100 °C.

If you bake your vacuum system and the temperature of the system exceeds 200 °C, you must put a radiation shield between the system and the EXT pump. This radiation shield will reduce the heat radiated onto the pump rotor.

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Typically, a bakeout of four hours is long enough to remove water condensation from the pump. However, the bakeout time will depend on the amount of condensation in the pump and the vacuum system, and the ultimate pressure you want to achieve.

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5 Maintenance

WARNING



Allow the pump-rotor to stop, then disconnect the EXC Controller before you remove the pump from your vacuum system for maintenance or fault-finding procedures.

5.1 Introduction

The maintenance operations for the EXT Turbomolecular pumps are described in the following sections. The ISX inlet-screen, the WCX water-cooler, the inlet-strainer, interstage-port seal and inlet-flange seals are available as spares (refer to Section 7). Fit the ISX inlet-screen as described in Section 3.3.1. Fit the WCX water-cooler as described in Section 3.7.2.

5.2 Bearing life

When supplied, the pump contains sufficient lubricant to supply the bearings for life. No routine maintenance is therefore required between bearing replacements. The bearings are not user-serviceable. The bearings will need to be replaced when they reach the end of their service life. This is typically more than 20,000 hours, but may be less; this depends on the type of pumping duty on which the pump is used.

When the bearings need replacement, we recommend that you exchange your pump for a factory reconditioned replacement. Alternatively, you can send your pump to an Edwards Service Centre to have the bearings replaced.

When you return EXT pumps to Edwards Service Centres please use the procedure included at the end of this manual. However, the instruction to drain all fluids does not apply to the lubricant in the EXT pump oil-reservoirs.

5.3 Rotor life

The life of the EXT pump rotor is typically 40,000 to 50,000 cycles (of acceleration to full speed, and then deceleration to a stop). The pump rotor is not user-serviceable.

We therefore recommend that you exchange your pump for a factory reconditioned replacement every 20,000 cycles, or 10 years of use, whichever occurs first. Alternatively, you can send your pump to an Edwards Service Centre for a major service (which will include rotor replacement).

When you return EXT pumps to Edwards Service Centres please obey the procedure included at the end of this manual. However, the instruction to drain all fluids does not apply to the lubricant in the EXT pump oil-reservoirs.

5.4 Clean the pump

WARNING



Clean the external surfaces of the EXT pump in a well-ventilated location. When you use cleaning solutions and solvents to clean the pump, observe all precautions specified by the manufacturer. Avoid inhalation of any particulates which may be present in the pump.

CAUTION

Do not attempt to clean any parts of the EXT pump other than the external surfaces. Organic solvents may damage internal pump components. Do not use abrasive materials to clean any part of the pump.

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If the inside of the EXT pump is contaminated, it may not be possible to achieve the specified ultimate vacuum, or pump-down time may increase. In these circumstances, you should return the pump to an Edwards Service Centre, where the pump will be dismantled and cleaned. Use the procedure given in the forms at the end of this manual to return the pump.

You can use any organic solvent to clean the external surfaces of the EXT pump. We recommend that you use non-CFC solvents, such as isopropanol or ethanol. Use a cleaning solution which is suitable for the contaminants on the pump surfaces.

For environmental reasons, keep wastage of cleaning solutions and solvents to a minimum.

5.5 Fault finding

Refer to Table 4 for the possible causes of faults and for the recommended actions to rectify faults. Table 4 is applicable to a basic, manually controlled pumping system with an EXC Controller configured for local (manual) operation.

Note that if you use an EXDC Pump Drive Module to control the EXT pump, or if you use an EXC Controller configured for remote operation to control the EXT pump, some of the checks and actions in Table 4 may not apply to your system.

Refer to the fault finding section of the instruction manual supplied with your EXDC Pump Drive Module or EXC Controller for further fault finding information.

Table 4 - Fault finding

Symptom	Check	Action
The pump does not rotate. After pressing start - Fail LED not lit.	Is the EXC Controller power LED lit?	If not, check that the electrical supply is on, check that the switch at the rear of the EXC Controller is on, check the fuse in the rear of the EXC Controller. If all of the above are OK then the EXC Controller is faulty. Consult Edwards or your supplier.
	Is the EXC Controller Start/Stop LED flashing?	If so, check that the correct links are made on the EXC Controller logic interface (refer to the instruction manual supplied with the EXC Controller). Check that any system interlocks are correctly made (refer to the instruction manual supplied with the EXC Controller). Check that the pump-to-controller cable is connected. If you have made all of the above checks and cannot identify the cause of the fault, consult Edwards or your supplier.
	Is the EXC Controller first speed indication LED lit?	If not, the EXC Controller is faulty. If lit, then the EXT pump is faulty. Consult Edwards or your supplier.
The EXC Controller trips into Fail - at any speed.	Are the system interlocks correctly connected?	Ensure that the system interlocks do not open after the EXT pump has started.

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Table 4 - Fault finding (continued)

Symptom	Check	Action
The EXC Controller trips into Fail during the ramp-up and before 50% speed is reached.	Is the inlet pressure too high? Is the backing pressure too high?	If so, reduce the pumping load, or check for a gross leak into the system.
	Is the EXT pump running too hot?	Increase the cooling-water flow or decrease the water temperature or do both. You may need to change from air-cooling to water-cooling. (Refer to Section 2 for maximum inlet pressure and cooling requirements). Check that external heat sources (such as system bakeout heaters) are not excessive.
	Does the rotor rotate freely?	If not, the EXT pump-bearings are damaged. Consult Edwards or your supplier.
	Is the timer set incorrectly?	Increase the timer setting (refer to the instruction manual supplied with the EXC Controller). If the EXC Controller still trips into Fail consult Edwards or your supplier.
The EXC Controller trips into Fail after 50% speed has been reached - the first two speed LEDs are lit.	Is the pressure too high?	If so, reduce the pumping load or check for a gross leak into the system.
		If the high gas load is temporary, configure the EXC Controller to delay the Fail trip on 50% speed and set an appropriate delay time (refer to the instruction manual supplied with the EXC Controller).
	Is the EXT pump running too hot?	Increase the cooling-water flow or decrease the water temperature or do both. You may need to change from air-cooling to water-cooling.
	Does the EXT pump rotor rotate freely?	If not, the EXT pump-bearings are damaged. Consult Edwards or your supplier.
The EXC Controller trips into Fail - all the speed LEDs are lit.	-	Consult Edwards or your supplier.

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Table 4 - Fault finding (continued)

Symptom	Check	Action
Ultimate pressure cannot be reached.	Is the pressure limited by water vapour ?	Bake the system and pump.
	Are any of the vacuum gauges contaminated ?	If so, clean or replace them.
	Is the pumping speed insufficient (due to poor conductance between the pump and the gauge or too large a chamber) ?	Increase the conductance or reduce the volume.
	Is the interstage inlet pressure > 0.2 mbar (20 Pa) ?	If the interstage inlet pressure is too high, inlet pressure at the turbomolecular inlet is reduced; ensure that the interstage inlet pressure is < 0.2 mbar (20 Pa).
	Is the backing pressure < 12 mbar (1200 Pa)?	Check for backing line leaks. If the backing pressure is too high, you may need a larger backing-pump.
	Is the high-vacuum area of the system contaminated ?	If so, clean the high-vacuum system.
	Check the rest of your system for leaks and contamination.	If found, clean the contaminated areas and repair the leaks.
	Remove the pump from the system and test the ultimate pressure of the pump alone (see Section 2 for specification).	If poor, check the pump for contamination and if necessary return the pump as described in Section 5.4. Leak-check the pump. If the leak rate > 1×10^{-7} mbar l s^{-1} (1×10^{-5} Pa l s^{-1}), consult Edwards or your supplier.
The EXT is very noisy or there is excessive vibration or both.	Is the pump rotational speed the same as the resonant frequency of the attached system ?	If so, change the natural frequency of your system or isolate the pump using flexible bellows.
	Is the vibration being transmitted from the rotary pump ?	If so, fit flexible bellows or a vibration isolator in the backing line.
	Is the noise irregular and getting progressively worse ?	If so, a bearing is defective. Consult Edwards or your supplier.
	Is the EXT making a constant high-pitched noise ?	If so, the rotor is out of balance. Consult Edwards or your supplier.
None of the above.	-	Consult Edwards or your supplier.

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6 Storage and disposal

6.1 Storage

Use the following procedure to store the pump.

1. Place protective covers over the inlet, outlet, interstage (EXT255Hi/EXT255HVi only), purge and vent ports.
2. Place the pump in its packing materials. For fastest pump-down when the pump is put back into service, seal the pump inside a plastic bag together with a suitable desiccant.
3. Store the pump in cool, dry conditions until required for use. When required, prepare and install the pump as described in Section 3.
4. Keep the pump upright at all times to prevent the drainage of oil from the bearing reservoir.
5. Avoid long-term storage if possible. When long-term storage is necessary, the pump should be set up and run for at least eight hours every six months.

6.2 Disposal

WARNING



In the unlikely event that the rotor fails, dust may be generated from the carbon fibre reinforced components in the pump. If the rotor has failed:

- **Wear the appropriate personal protective equipment (gloves, a dust-proof mask which covers the nose and mouth, an apron or overalls, and safety glasses or goggles) when you handle the pump.**
- **Ensure that the pump inlet, outlet, and vent-, interstage- and purge-ports are all blanked off/sealed before you dispose of the EXT pump.**

Dispose of the EXT Turbomolecular Pump and any components and accessories safely in accordance with all local and national safety and environmental requirements.

Take particular care with any components which have been contaminated with dangerous process substances.

Take appropriate action to avoid inhalation of any particulates which may be present in the pump.

Do not incinerate the pump. The pump contains phenolic and fluorosilicone materials which can decompose to very dangerous substances when heated to high temperatures.

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7 Service, spares and accessories

7.1 Introduction

Edwards products, spares and accessories are available from Edwards companies in Belgium, Brazil, China, France, Germany, Israel, Italy, Japan, Korea, Singapore, United Kingdom, USA and a world-wide network of distributors. The majority of these employ service engineers who have undergone comprehensive Edwards training courses.

Order spare parts and accessories from your nearest Edwards company or distributor. When you order, please state for each part required:

- Model and Item Number of your equipment
- Serial number (if any)
- Item Number and description of the part

7.2 Service

Edwards products are supported by a world-wide network of Edwards Service Centres. Each Service Centre offers a wide range of options including: equipment decontamination; service exchange; repair; rebuild and testing to factory specifications. Equipment which has been serviced, repaired or rebuilt is returned with a full warranty.

Your local Service Centre can also provide Edwards engineers to support on-site maintenance, service or repair of your equipment.

For more information about service options, contact your nearest Service Centre or other Edwards company.

7.3 Spares

7.3.1 ISX inlet-screen

An inlet-screen is fitted to your pump as supplied to prevent damage from the entry of debris into the pump. The Item Numbers of replacement inlet-screens are given below. Select the inlet-screen according to the pump inlet-flange size.

Flange size	Inlet-screen	Item Number
DN100ISO-K/DN100CF	ISX100	B580-51-001

7.3.2 Inlet-strainer (EXT255Hi pump only)

The EXT255Hi pump is supplied with an inlet-strainer for the interstage-port. The Item Number of a replacement inlet-strainer is given below.

Interstage-port size	Item Number
DN25NW	A223-06-067

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7.3.3 Interstage-port seal (EXT255HVi pump only)

The EXT255HVi pump is supplied with a fluoroelastomer 'O' ring seal for the interstage-port. The Item Number of a replacement interstage-port seal is given below.

Interstage-port size	Item Number
DN40NW	C105-16-395

7.3.4 WCX water-cooler

A water-cooler can be fitted to the EXT pump. Please refer to Section 3 to check the suitability of water cooling for a particular application. Pumps with a Conflat flange are supplied with a water-cooler.

Water-cooler	Item Number
WCX500	B580-61-001

7.3.5 Inlet-flange seals

EXT pumps are supplied with an inlet seal. The Item Numbers of replacement seals are given below.

Inlet-flange	Inlet seal	Item Number
DN100ISO-K	ISO100 trapped 'O' ring, fluoroelastomer	B271-58-171
DN100CF	Copper compression gasket (pack of 5)	C082-00-008

7.4 Accessories

7.4.1 Installation

The accessories available for use with the EXT turbomolecular pumps are described in the following Sections. Figure 9 shows how the accessories are fitted to an EXT pump.

7.4.2 EXC controller

The Edwards EXC Controllers provide the facilities necessary for operating a pumping system based on an EXT255H, EXT255Hi or EXT255HVi pump. The Item Numbers of the EXC Controllers are given below.

Controller	Voltage	Item Number
EXC100E	90 - 264 V a.c.	D396-20-000
EXC100L	90 - 264 V a.c.	D396-22-000
EXC120	90 - 264 V a.c.	D396-16-000
EXC300	90 - 132/180 - 264 V a.c.	D396-14-000

7.4.3 Pump-to-controller cable

You must fit a pump-to-controller cable between an EXC Controller and the EXT pump. A cable is not supplied with the EXT Pump or the EXC Controller (except EXC100L). The following cables are available:

Cable	Length	Item Number
Pump-to-controller	1 m	D396-18-010
Pump-to-controller	3 m	D396-18-030
Pump-to-controller	5 m	D396-18-050

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7.4.4 EXDC drive modules

Fit an EXDC Drive Module as an alternative to an EXC Controller and pump-to-controller cable.

Drive Module	Item Number
EXDC80	D396-40-000
EXDC160	D396-41-000

7.4.5 BX bakeout band

A BX bakeout band accelerates the degassing of the pump to enable it to achieve lower pressures. It may also be used to protect the pump from condensation of contaminants. The bakeout bands are available in 110-120 V or 220-240 V versions and may be powered from a rear panel socket on the EXC Controller.

Pump	Bakeout band	Item Number
EXT255H/Hi	BX250 (110 V)	B580-52-041
	BX250 (240 V)	B580-52-061

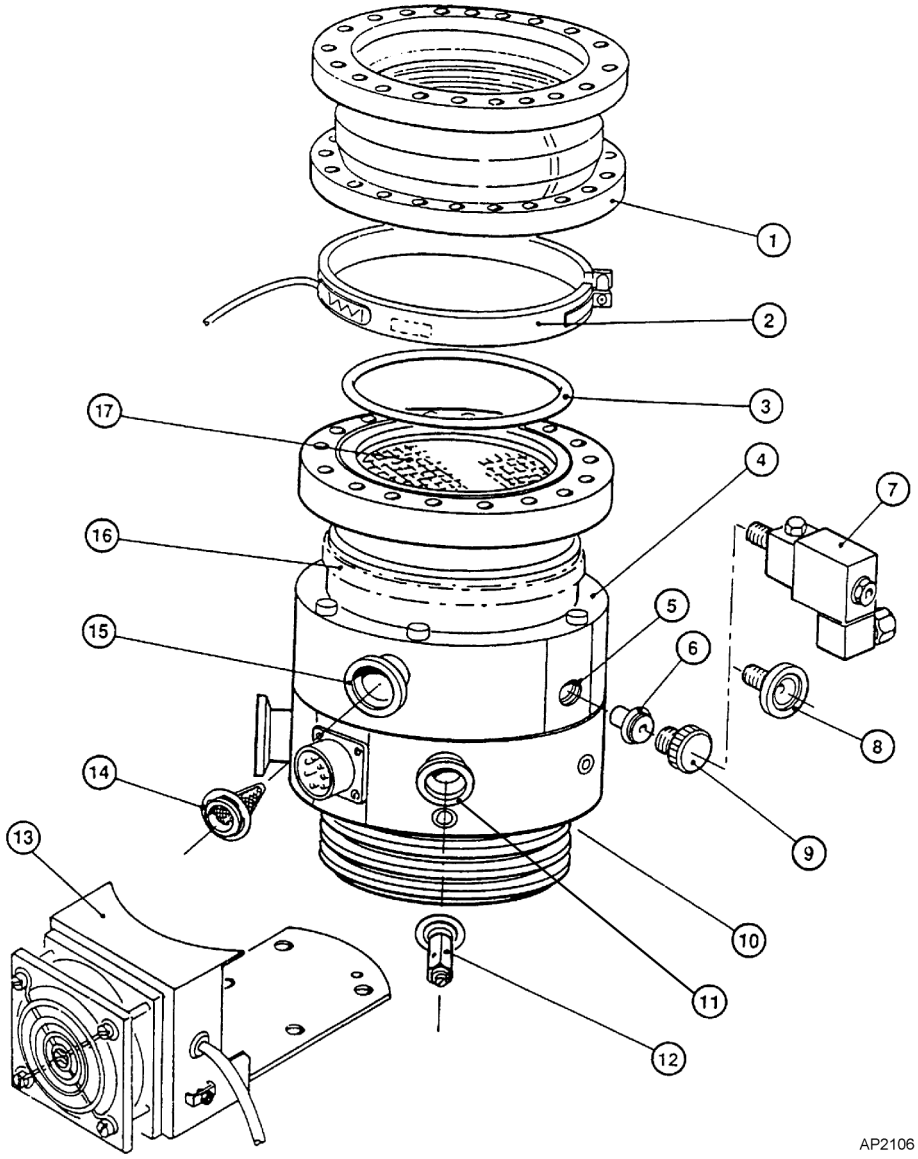
7.4.6 FL20K foreline trap

The foreline trap minimises oil vapour backstreaming from the backing pump and is recommended where the highest system cleanliness is required.

Foreline trap	Item Number
FL20K	A133-05-000

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Figure 9 - Installation of optional accessories (and spares): EXT255Hi shown



1. Vibration isolator
2. Bakeout band
3. Inlet-flange seal (supplied)
4. EXT255Hi pump
5. Vent-port
6. VRX vent-restrictor
7. Solenoid vent-valve
8. DN10NW adaptor
9. Manual vent-valve (supplied)
10. Water-cooler (on opposite side of the pump)
11. Vent port adaptor
12. Purge restrictor
13. Air-cooler
14. Inlet strainer (supplied)*
15. Interstage-port†
16. Bakeout band position
17. Inlet-screen (supplied)

* *EXT255Hi only*
† *EXT255Hi/EXT255HVi only*

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7.4.7 TAV vent-valve and vent-port adaptor

Two solenoid-operated vent-valves are available for system venting. The valves are 24 V d.c., normally-open, and can be driven automatically from the EXC Controller. The solenoid-valve is fitted in place of the manual-valve, or alternatively can be fitted with an adaptor (supplied with the valve) and be used with any suitable NW10 flanged port on your vacuum system.

The vent-port adaptor allows the vent-port or the purge-port to be used with any suitable NW10 fitting: see Figure 9, item 8 and item 11.

Product	Item Number
TAV5 vent-valve	B580-66-010
TAV6 vent-valve	B580-66-020
NW10-1/8 inch BSP male adaptor	B580-66-011

7.4.8 ACX air-cooler

An ACX air-cooler can be fitted to the EXT pump. However, please refer to Section 3 to check the suitability of air-cooling in a particular application.

Pump	Air-cooler	Item Number
EXT255H/Hi	ACX250H	B580-53-160

7.4.9 Vibration isolators

In applications where the small amount of vibration generated by the turbomolecular pump is a problem, a vibration isolator can be fitted. The isolator consists of two special flanges separated by a flexible bellows and a rubber, anti-vibration, outer collar. The isolator required depends on the pump inlet-flange size.

Inlet-flange size	Item Number
DN100ISO-K	B580-20-000

7.4.10 PRX purge-restrictor

A modified DN10NW centring-ring is available to filter the purge gas and restrict its flow rate to the recommended flow of 25 sccm. The restrictor is suitable for all EXT pumps fitted with a purge-port.

Purge-restrictor	Flange size	Item Number
PRX10	NW10	B580-65-001

7.4.11 VRX vent-restrictor

Use a VRX fixed orifice vent-restrictor to restrict the flow of vent gas into the EXT pump. Refer to Section 3.4 for information on the selection of the correct VRX vent-restrictor.

Vent-restrictor	Orifice diameter (mm)	Item Number
VRX10	0.1	B580-66-021
VRX20	0.2	B580-66-022
VRX30	0.3	B580-66-023
VRX50	0.5	B580-66-024
VRX70	0.7	B580-66-025

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